# Prediction Presupposes Persistence

Margaret Stokes

*ORCID: 0009-0004-6422-4174*

*July 2026*

## 1. The World Comes First

There is a world. People arrive in it. Nobody consults them about this; they simply find themselves here, among things that were here before them and will remain after them.

For a while, perhaps, a person could live in that world alone. But the moment anyone wants to share something — tell someone where they live, warn them about the river, teach them how to start a fire — something has to be true that we almost never say aloud, because it is too obvious to say: **the world has to still be there.**

Think about what "how to start a fire" actually requires. It requires that fire behaves tomorrow as it behaved yesterday. That dry wood is still dry wood when the listener goes looking for it. That the person receiving the instruction persists long enough to use it, in a world that has persisted long enough to be used. The instruction is a promise drawn against a world that must remain in place to honour it. Take the world away — or let it flicker, reset, or exist only while someone is looking at it — and the instruction is not merely useless. It is meaningless. There is nothing for the words to be *about*.

This gives us the order of things, and the order is not negotiable:

> *First, a world that persists.*
> *Then, people in it.*
> *Then, the desire to share it.*
> *Then — and only then — language.*

Language is not the foundation. Language is the *technology of sharing a persistent world*. It came last because it depends on everything before it. No world, no people. No people, no need to share. Nothing to share, no language. Ipso facto: the world comes first.

This is so basic that it sounds like it could not possibly matter. But the systems we now build to handle language are built in the reverse order. They begin with language — oceans of it — and attempt to conjure a world out of it, or lately, to infer a world from streams of observation, assembled fresh with every prediction. The world, in these systems, is an *output*: something derived, reconstructed, guessed at. It exists only while the machine is predicting it.

That is the roof before the ground.

The claim of this paper is simple, and it is falsifiable: **any system that produces language without a persistent world beneath it will fail in exactly the places where language depends on persistence** — in referring to the same thing twice, in knowing that a thing continues to exist between mentions, in saying "she" and meaning someone in particular, and in the most important act of all: refusing to speak when there is nothing there to speak about.

The rest of this paper says why, names where those failures will appear, and shows that the field has already conceded half the argument without noticing it has done so.

## 2. The Half-Concession

For most of the past decade, the field's working assumption was that language could come first. Feed a system enough text and a world would condense out of it — or near enough to one that nobody would check. Meaning was treated as a property of words, extractable from words alone. The whole apparatus — the training objectives, the benchmarks, the interfaces — encodes a single quiet assumption: *extract(text)*. Text in, meaning out, no world required.

That assumption is now collapsing in public, and the collapse is expensive.

In late 2025 and through 2026, the industry began a loud pivot toward what it calls *world models*: systems meant to learn from reality rather than from text about reality. The most prominent voice in this turn, Yann LeCun, left one of the largest AI companies on earth to found a venture on precisely this premise, raising over a billion dollars before shipping a product. His stated reason deserves quoting for what it admits: next-word prediction is architecturally insufficient because such systems learn no model of how events cause one another. His company's founding pitch is that machines should learn from reality, not just from language. His CEO arrived at the same conclusion through medicine, where a system that speaks without a world behind its words can kill someone.

Set aside the valuations and hear what is actually being said. The field's most decorated researchers, backed by its most serious money, are now asserting: *language alone is not enough; there must be a world.*

That is the first half of this paper's argument, conceded — publicly, expensively, and by the people best positioned to have denied it.

But it is only half. Because look at where the world sits in these new systems. It is *inferred*. It is reconstructed from streams of observation, assembled by prediction, held in latent variables that exist while the model runs and in the form the model recovers them. The mathematics being offered as foundations confirms this: the formal guarantees are about how faithfully hidden structure can be *recovered from observation* — and even those guarantees hold only under conditions (well-behaved distributions, stationary dynamics) that reality declines to satisfy. Nowhere in this apparatus is there a commitment to things that *continue to exist between observations*. Nowhere is there an entity that persists whether or not it is currently being predicted.

In these systems, the world is still an output. A better output — grounded in observation rather than conjured from text — but an output all the same. Existence is derived from prediction. The dependency still runs backwards.

The concession, then, is exactly half-made. The field has agreed that language needs ground beneath it. It has not yet noticed that ground which is generated on demand is not ground. A world that exists only while the machine predicts it cannot honour the promise that every instruction, every reference, every shared word draws against it — that it will *still be there*.

They have agreed the roof needs something under it. They are now building a very good picture of the ground, and preparing to stand on the picture.

## 3. Looking in the Wrong Place

It is worth being fair about what the world-models programme gets right, because the disagreement is sharper for it. Learning from reality rather than from text is a genuine correction. A system with good causal representations will make fewer absurd claims than one that has only ever seen words. If the problem were *competence* — how good the machine's picture of the world is — this programme would be the answer, and better pictures would eventually be good enough.

But the problem was never competence. The problem is *direction*.

Ask a simple question of any of these systems: does the cup on the table exist when the model is not predicting it?

In a prediction-first architecture, the honest answer is no — or rather, the question cannot be asked. The cup is a latent variable, recovered from observation, held for as long as the inference runs, reconstructed afresh the next time it is needed. Between predictions there is no cup. There is no *between*, because there is nothing that persists across it. The world in these systems is like a film: convincing frame by frame, with nothing continuing underneath from one frame to the next except the machinery of producing frames.

Now ask what language needs, and put it against that.

Language needs the cup to be *the same cup* tomorrow. When I say "the cup I gave you," those words reach for something that has existed continuously since the giving — not something re-derived from fresh observation and assumed, hopefully, to be the same. When I say "she," the word only works because there is a particular person who has persisted in our shared frame since I first mentioned her. When I teach you to start a fire, the teaching draws on a world that holds still between the lesson and the doing. Every one of these acts presupposes existence that does not depend on anyone currently looking, predicting, or inferring.

This gives us two architectures, and they differ in one thing only — which way the dependency runs:

In the prediction-first architecture, **existence is derived from prediction.** The world is whatever the model currently recovers. Things exist because they are inferred.

In the persistence-first architecture, **prediction is licensed by existence.** The world is prior. Things are inferred because they exist, and an inference about a thing that is not in the world has no standing at all — however confident it sounds.

That last clause is where the difference stops being philosophy and becomes engineering. A prediction-first system has no principled way to refuse. If the world is whatever the model recovers, then whatever the model recovers *is* the world, and every output arrives with the same credentials. There is no independent fact of the matter against which an output could be found illegitimate. Confidence becomes the only currency, and confidence is exactly what these systems counterfeit best. But a persistence-first system can say something no amount of predictive skill can say: *there is nothing there*. The referent does not exist in the frame. The question has no standing. I will not guess. Refusal — real refusal, grounded refusal — is only possible when there is a world to be wrong about.

So the objection to the world-models programme is not that it is building badly. It is building well, in the wrong place. It is improving the machine's picture of the ground, when the question was never the quality of the picture. The question is whether there is ground — something prior, persistent, and independent of the predicting — for the picture to be a picture *of*. No refinement of inference closes that gap, because the gap is not in the inference. It is in the order of things. And the order, as section 1 said, is not negotiable.

## 4. The Tree in the Forest

There is an old objection waiting here, and it should be met head-on rather than stepped around.

Epistemology, the objection runs, forbids the claim. You cannot *prove* the cup continues to exist while nobody perceives it. If a tree falls in a forest with no one to hear, who can say it makes a sound? Perhaps to be is to be perceived, and the world really is only what shows up in observation. On this view, the prediction-first architecture is not making an error at all — it is simply being honest about the limits of knowledge. The world *as recovered from observation* is the only world anyone ever gets.

Three answers, in ascending order of force.

First, the historical one. This objection is not new; it is Bishop Berkeley's, three centuries old — *esse est percipi*, to be is to be perceived. And it is worth noticing what the prediction-first architecture actually amounts to: Berkeley rebuilt in silicon. To be is to be predicted. The cup exists while inferred; the forest exists while observed. But Berkeley understood that his position collapsed unless *something* perceived continuously — his answer was God, whose unbroken attention kept the quad in existence between the students' glances. The world-models programme has adopted Berkeley's metaphysics and omitted his load-bearing part. It has idealism without the divine observer: a world that genuinely blinks out between inference cycles, with nothing minding the store. That is not epistemic honesty. It is an incoherence Berkeley himself already diagnosed and patched.

Second, the plain one. The tree makes a sound. Air is compressed whether or not an ear is present; the wave propagates; the physics does not check for an audience. What the objection actually shows is a confusion between two different questions — *what exists* and *what can be proven to exist from where I stand*. Epistemology governs the second. It has no authority over the first. The limits of my knowledge are facts about me, not facts about the forest.

Third, and decisively: the skeptic refutes themselves in the asking. To demand "prove that *the cup* continues to exist," the skeptic must say "the cup" — a referring expression, reaching for a particular thing that persists from their sentence to my reply. The demand for proof is issued *in language*. And language, as section 1 established, only works over a persistent world. The question presupposes the very thing it purports to doubt. One cannot coherently use the technology of sharing a persistent world to deny that the world persists. Anyone who asks the tree question in words has already conceded the answer.

So persistence is not a conclusion that owes anyone a proof. It is a precondition of the conversation in which proof could be demanded — which is precisely why it must be built in rather than inferred on the fly.

It is worth naming what happens when the objection wins anyway. Doubt of this kind does not sharpen inquiry; it produces a *skeptical inertia* — a doubt that functions as a parking brake rather than a whetstone. And beneath it sits a strange arrogance wearing humility's coat: the quiet promotion of the observer to a load-bearing element of the universe, as though things stop happening because I cannot see them. The limits of my knowledge are facts about me. They are not instructions to the forest, which was running long before anything evolved ears and has managed continuity without supervision ever since. Real humility runs the other way: the world does not depend on my attention, and no architecture should be built as though it does.

## 5. What Language Already Knows

None of this is speculation about language. It is observation of it. Look closely at the machinery of ordinary talk and you find persistence assumed at every joint — not as a philosophical garnish but as the working mechanism, the thing without which the simplest sentences fail.

Start with the smallest words, because they carry the heaviest load.

**"She."** Three letters, and they only work if a particular person, mentioned earlier, has continued to exist in our shared frame from that mention to this one. The word does not describe her; it *reaches* for her — and reaching requires that she still be there to reach. Say "my daughter called this morning; she sounded well," and the *she* travels backwards across the sentence boundary to a person who must have persisted across it. If nothing persists between sentences, "she" refers to no one, and half of every conversation ever held dissolves.

**"The."** We say "the cup," not "a cup," precisely when the cup is already established between us — when it has been entered into our shared frame and is presumed to still be sitting there. The definite article is a persistence claim compressed into three letters: *you know the one; it's still there.*

**"Still," "again," "back," "the same."** An entire family of words whose only job is to track existence across time. "The fire went out; light it again." *Again* is meaningless unless the fire that went out and the fire to be lit stand in a relation across a gap of time — unless there is a world in which the gap occurred.

Now, I arrived at this by looking — at language, at people, at what sharing a world actually requires. It turns out that formal linguistics spent some forty years working the same ground by a longer road, and the convergence is worth having on record.[1] From the early 1980s onward, semanticists studying exactly these puzzles — how "she" finds its referent, how "the" differs from "a," how meaning accumulates across sentences — were forced to a structural conclusion: the meaning of a sentence cannot be computed from the sentence alone. They found they had to model conversation as building and maintaining a running registry of things — picture a growing stack of file cards, one per entity, opened when a thing is first mentioned, updated as talk continues, *persisting between sentences* so that later words can reach back to earlier things. On this account, the meaning of an utterance is not a property of its words. It is the *change* the utterance makes to the registry. Meaning is an update to a persistent record — and without the record, there is nothing to update, and so nothing meant.

They were driven to this not by metaphysical taste but by the data. Sentences that are perfectly interpretable in sequence become uninterpretable in isolation. The persistence layer was not an optional enrichment; it was the only way the machinery of reference could be made to work at all.

All of this compresses into one line, and the line is the centrepiece of this paper. The prevailing architecture of language systems assumes that meaning has this shape:

**`extract(text)`**

Text in, meaning out. But four decades of formal semantics, three centuries of philosophy, and one afternoon of watching what "she" actually does all converge on the correction:

**`extract(text, world)`**

There is no meaning function whose domain is text alone. The second argument is not an enhancement, an add-on, or a context window. It is the persistent world against which every referring word cashes out — and dropping it is not a simplification of the function. It is a different function, one that computes something meaning-shaped from half of meaning's inputs, and the missing half is the half that made it *about* anything.

Every system built on the first signature is running on borrowed coherence — inheriting, from its training text, the shadow of worlds that persisted for the humans who wrote it, while maintaining no world of its own. The shadow is often convincing. It is still a shadow. And the places where shadows fail are predictable, which is the subject of the next section.

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[1] The convergence with formal work on discourse representation and file-change semantics from the early 1980s (Kamp; Heim) is noted here as corroboration. No claim of lineage is made in either direction; the agreement is the point, and the agreement is checkable.

## 6. Where the Shadows Fail

A position is only worth holding if it can be wrong. So here is where this one stakes itself: if the argument of this paper is correct, then systems without a persistent world beneath their language — however sophisticated their prediction, however grounded their training — will keep failing in the following specific places. These are not vague warnings. They are the load points of language identified in section 5, and each one names an observable failure.

**First: reference will drift.** Over a long enough exchange, "she," "it," "the earlier one" will detach from their referents and reattach to the wrong things — because in a system where every mention re-derives its object rather than reaching for a persisting one, nothing anchors the third mention to the first. Two entities discussed at length will bleed into each other; properties of one will migrate to the other. The longer the discourse, the worse the drift, and no improvement in per-step prediction will cure it, because the failure is not in the steps. It is in the absence of anything between them.

**Second: identity will not hold across time.** A thing established early — a constraint, a fact, a decision — will silently mutate by the end, not through contradiction the system notices, but through contradiction it *cannot* notice, because noticing requires comparing the current claim against a persisting record, and there is no record; there is only the fresh inference. The words "still," "again," and "the same" will be produced fluently and honoured erratically.

**Third: absence will be invisible.** This is the deepest one. A persistence-first system can represent the fact that something is *not in the world* — absence is a state of the record. A prediction-first system cannot; it can only fail to retrieve, and a failed retrieval looks, from the inside, exactly like a prompt for more inference. So when asked about a thing that does not exist, the system will not report the absence. It will *manufacture the referent* — a plausible citation, a person, a legal case, an API that was never written. The field calls this hallucination and treats it as a data problem, a training problem, a scale problem. This paper's claim is that it is none of those. It is an ontological problem: the confabulation of referents is what language use looks like when there is no world for the words to be about. It will shrink with scale. It will not go to zero, because nothing is being made *false* — there is no world in these systems for a claim to be false *against*.

**Fourth: refusal will remain ungrounded.** These systems will decline to answer when trained to decline, on patterns of when declining is expected — which is imitation of refusal, not refusal. Grounded refusal — *I will not answer because the referent is not in the frame; there is nothing there* — requires a frame for the referent to be absent from. Expect refusal behaviour to stay brittle, over-firing on the harmless and under-firing on the confidently fabricated, because it is keyed to the shape of questions rather than the state of a world.

**And a fifth prediction, about the field rather than the systems.** As these failures accumulate, engineers will patch them — with retrieval stores, memory modules, scratchpads, state trackers, entity registries bolted alongside the model. The author has watched this cycle from close range: demonstrate a failure publicly, and it is patched with impressive speed — but always the *instance*, never the *class*. The pronoun case gets fixed; the architecture that guarantees the next pronoun case remains untouched, because fixing the class would mean conceding the diagnosis. Each patch will work precisely to the degree that it does one thing: maintain something that persists between inferences, against which outputs can be checked. That is to say: the field will rebuild the persistent world piecemeal, under a dozen names, patch by patch, without ever conceding the principle — the way a builder who refuses to believe in foundations will nonetheless, brick by brick, end up putting one under a sagging house. Watch for it. Every successful mitigation will be a persistence commitment made locally and disavowed globally — a persistence layer wearing a different hat.

**What would prove this paper wrong** is equally specific: a system with no persistent world layer — pure prediction, however architected — that maintains stable reference over arbitrarily long discourse, holds identity across time, reliably reports absence instead of confabulating referents, and refuses on the basis of what is not there. If scale or representational quality alone delivers that, then persistence was emergent rather than prior, and the argument fails. I do not expect to be proven wrong, for the reason the whole paper gives: those capacities are not behaviours that sit *on top of* a world. They are what having a world *consists of*. A system that exhibits all of them has a persistent world by definition — whether or not its builders have noticed.

## 7. Prior Work

This paper is not a proposal. It is the argument beneath an architecture that already exists, and the record of that architecture is public, dated, and cryptographically anchored.

The persistence-first position described here is implemented in **Aurora-Lens**, a deterministic governance system for language models built by the author as sole inventor, with origins in **November 2025** — before the world-models funding wave, and before the industry's public turn toward grounding. Its core construct is the **Persistent Existence Frame (PEF)**: a world that is prior to, and independent of, any inference run against it. In this architecture the signature correction of section 5 is not aspiration but interface: extraction is *extract(text, pef)* — a delta operation over a persistent world — and *extract(text)* is not a supported degenerate case but a rejected category error. Meaning is conditional on persistent existence; guessing without state is disallowed; refusal is a first-class outcome, grounded in the state of the frame rather than the shape of the question. The failure modes predicted in section 6 are, in this architecture, not risks to be mitigated but operations that cannot be expressed.

The record is as follows. The system's provenance is anchored in GPG-verified repository commits dating to November 2025, independently timestamped via OpenTimestamps and archived by Software Heritage. A consolidated prior art and precedence record, with supporting artifacts, is published on Zenodo under the author's ORCID (0009-0004-6422-4174), alongside the architecture's foundational papers on SSRN and OSF. A working demonstration is publicly released, and the implementation is distributed as a published package on PyPI under AGPL-3.0 dual licensing. Patent protection over the architecture's methods is on foot in Australia, with priority established by provisional filings from the same period.

One item in that record deserves particular mention here, because it converts a prediction of this paper into an exhibit. The first failure mode named in section 6 — reference drift, the detachment of pronouns from their referents in systems with nothing persisting between inferences — is not merely predicted. It is empirically demonstrated in the author's published case study on pronoun ambiguity resolution in transformer-based reasoning (DOI: 10.5281/zenodo.21123912), on the record before this paper was written. The smallest word in the language, failing exactly where the architecture says it must.

**The record:**

*Zenodo (under ORCID 0009-0004-6422-4174):*

– Epistemic Legitimacy as a Governance Layer for Large Language Models: Architecture and Implementation — DOI: 10.5281/zenodo.18653120; revised deposit DOI: 10.5281/zenodo.19504665

– Aurora-Lens Demo v1.0 (milarien/aurora-governor-demo) — DOI: 10.5281/zenodo.18705898

– Operational Alignment of Aurora-Lens with OECD Due Diligence Guidance for Responsible AI (2026) — DOI: 10.5281/zenodo.18719033

– Present-Centered Cognition Model (PCCM): A Cross-Substrate Framework for Understanding Consciousness, Presence, and Latent States in Humans and Machines — DOI: 10.5281/zenodo.18976303

– Architectural Response to Nippon Life v. OpenAI: Runtime Admissibility as the Missing Layer — DOI: 10.5281/zenodo.19212492

– Aurora-Lens: A Provider-Agnostic Governance Proxy for Large Language Models — DOI: 10.5281/zenodo.19879302

– Aurora-Lens and Related Architecture: A Prior Art and Precedence Record — DOI: 10.5281/zenodo.19881958

– Present-Centred Cognition Model — Working Documents, November 2025 — DOI: 10.5281/zenodo.20046124

– A Terminology Translation Layer for AI Governance Architectures — DOI: 10.5281/zenodo.20356020

– Empirical Demonstration: Architectural Limitations in Transformer-Based Reasoning — A Case Study in Pronoun Ambiguity Resolution — DOI: 10.5281/zenodo.21123912

– When a Basis Goes Stale: Routing Consequence Through Admissibility — DOI: 10.5281/zenodo.21124705

– Aurora-Lens: Proof and Precedence — DOI: 10.5281/zenodo.21397474

*SSRN:*

– Epistemic Legitimacy as a Governance Layer for Large Language Models: Architecture and Implementation — SSRN 6244239; DOI: 10.2139/ssrn.6244239

– Aurora-Lens and Related Architecture: A Prior Art and Precedence Record — SSRN 6674398; DOI: 10.2139/ssrn.6674398

*OSF:*

– Epistemic Legitimacy as a Governance Layer for Large Language Models — osf.io/86bxj

*GitHub (public repositories):*

– github.com/milarien/aurora-governor-demo

– github.com/milarien/aurora-evidence-dossier

– github.com/milarien/conceptual-bridges-archive

The point of this section is not decoration. It is the timestamp made explicit. The argument of this paper — that the world must come first, that language is the technology of sharing a persistent world, that any system producing language without one will fail at reference, identity, absence, and refusal — was not composed after the field's turn toward grounding, as commentary upon it. It was built, filed, and published before it, and the construct at its centre is named for the claim: *Persistent Existence*.

The field is welcome to arrive. The ground was here first. That is rather the point.

And this converts directly into engineering. A system doing language does not need metaphysical certainty that the world persists — no one has that, and no one needs it. It needs the *commitment*: a frame in which things, once established, continue to exist unless something changes them; in which identity holds between mentions; in which absence is a fact and not a gap in inference. Language makes that commitment in every sentence. An architecture that handles language while refusing the commitment is not being cautious. It is speaking a language whose one non-negotiable premise it declines to honour.